

Multiple Choice

1. **Answer:** C

Options: A) Bundle sheath, companion cell, phloem parenchyma, fibers; B) Phloem parenchyma, bundle sheath, fibers, companion cell; C) Bundle sheath, phloem parenchyma, companion cell, sieve tube; D) Bundle sheath, fibers, companion cell, sieve tube; E) Fibers, bundle sheath, companion cell, sieve tube

Note: Correct option: C - Bundle sheath, phloem parenchyma, companion cell, sieve tube

2. **Answer:** A

Options: A) The thick muscular wall of the stomach; B) Antibacterial properties of stomach acid; C) Gastric juice; D) High pH levels in the stomach; E) Presence of beneficial bacteria

Note: Correct option: A - The thick muscular wall of the stomach

3. **Answer:** A

Options: A) Shivering is a method the body uses to relieve pain.; B) Shivering is a tactic to ward off predators by appearing agitated.; C) Shivering is a way to burn excess calories; D) Shivering is a response to fear or anxiety; E) Shivering is an involuntary action to prepare muscles for physical activity.

Note: Correct option: A - Shivering is a method the body uses to relieve pain.

4. **Answer:** A

Options: A) Taste, wind direction, humidity; B) Atmospheric oxygen levels, plant growth, tidal patterns; C) Infrared radiation, sound frequency, barometric pressure; D) Moon phases, echo location, gravitational pull; E) Sunlight intensity, water current, soil composition

Note: Correct option: A - Taste, wind direction, humidity

5. **Answer:** A

Options: A) Without HCl, more bacteria would be killed in the stomach.; B) No HCl would lead to reduced mucus production in the stomach lining.; C) The absence of HCl would cause the stomach to digest its own tissues.; D) No HCl in the stomach would significantly increase the pH of the small intestine.; E) The absence of HCl would speed up the digestive process.

Note: Correct option: A - Without HCl, more bacteria would be killed in the stomach.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Greater anterior laxity of the uninjured knee was associated with poorer stability and functional outcomes after ACL reconstruction. Excessive anterior laxity of the uninjured knee thus appears to represent a risk factor for inferior outcomes.

7. **Answer:** False

Options: A) True; B) False

Note: The survey shows significant dissatisfaction amongst consultant radiologists with the current service, confirms a low number of paediatric radiologists taking on this work, and suggests the potential to increase numbers of radiology child abuse experts by 27% if given improved training and support. Appropriate service and education strategies should be implemented.

8. **Answer:** False

Options: A) True; B) False

Note: Early oral intake is possible after laparotomy and colorectal resection. Thus, the laparoscopic surgeon's claim of early tolerated oral intake may not be unique to laparoscopy.

9. **Answer:** True

Options: A) True; B) False

Note: A higher PV in PMR increases the risk of prolonged steroid therapy and late GCA. Female sex and particular HLA alleles may increase the risk of late GCA. Starting patients on >15 mg prednisolone is associated with a prolonged steroid duration.

10. **Answer:** False

Options: A) True; B) False

Note: The incidence of venous thrombosis during hospitalisation in a department of general internal medicine is low and does not justify prophylaxis in all internal patients. Cancer is a strong risk factor for hospital-acquired thrombosis in the medical ward. Further studies may answer the question as to whether thrombosis prophylaxis in this subgroup is feasible.

Long Form Response

11. **Answer:** Complex genes interaction, environmental control, long lifespan, small family sizes, ethical issues. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Random mating. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key